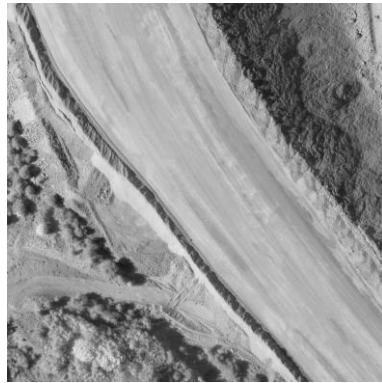
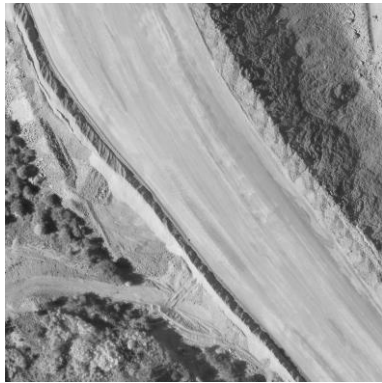
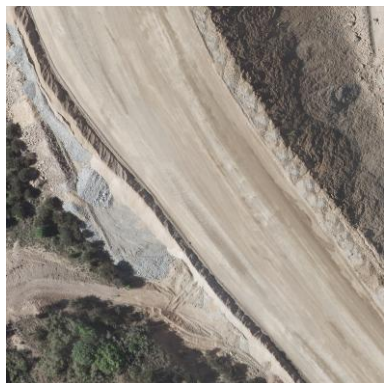


Input

Generated

Ground Truth

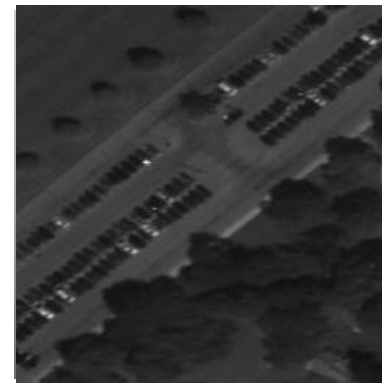
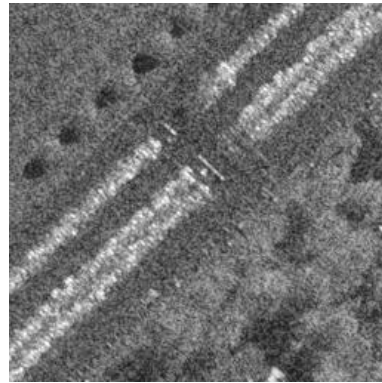


a) RGB  $\rightarrow$  IR

Input

Generated

Ground Truth

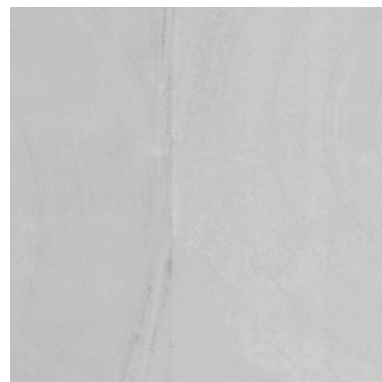
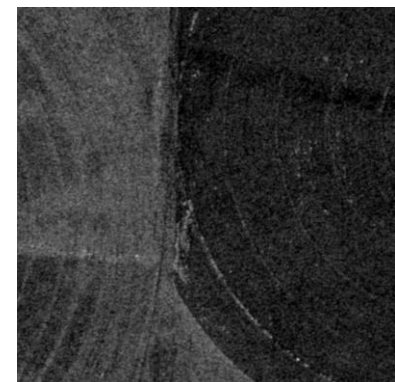


b) SAR  $\rightarrow$  EO

Input

Generated

Ground Truth



c) SAR  $\rightarrow$  IR

Input

Generated

Ground Truth



d) SAR  $\rightarrow$  RGB

# Reverse Bridge / DBIM Generation

$$t : T \rightarrow 0$$

Reverse Bridge

$$z_t = a_t x + b_t y + \sigma_t \epsilon$$

$x$

$y$

Source  
distribution

Target  
distribution

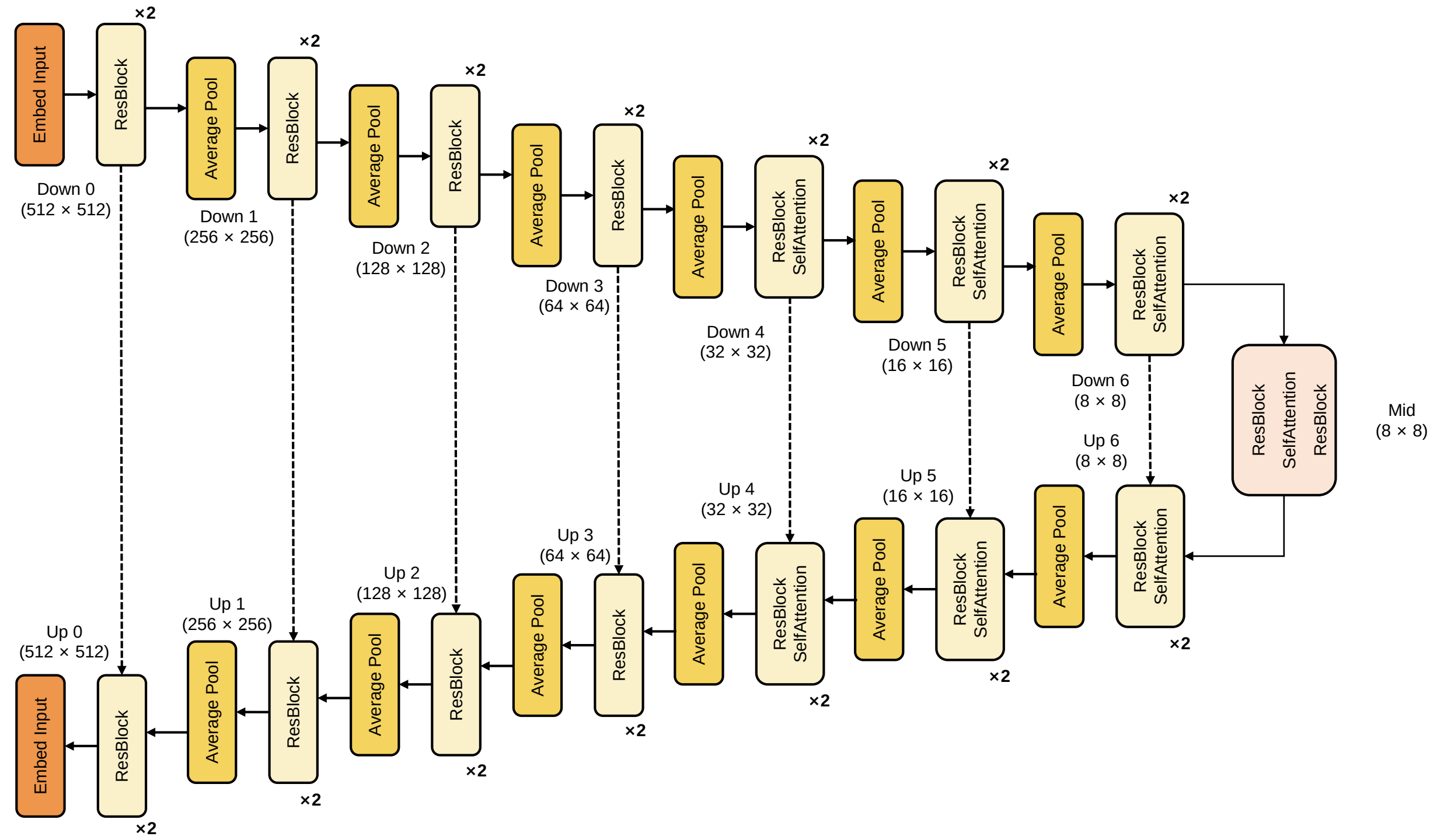
Place-  
holder

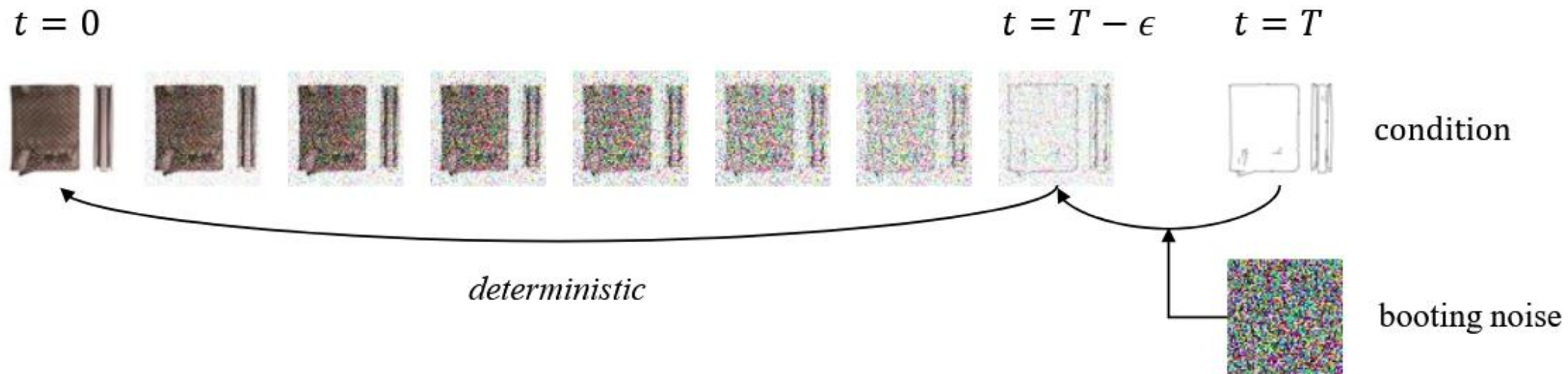
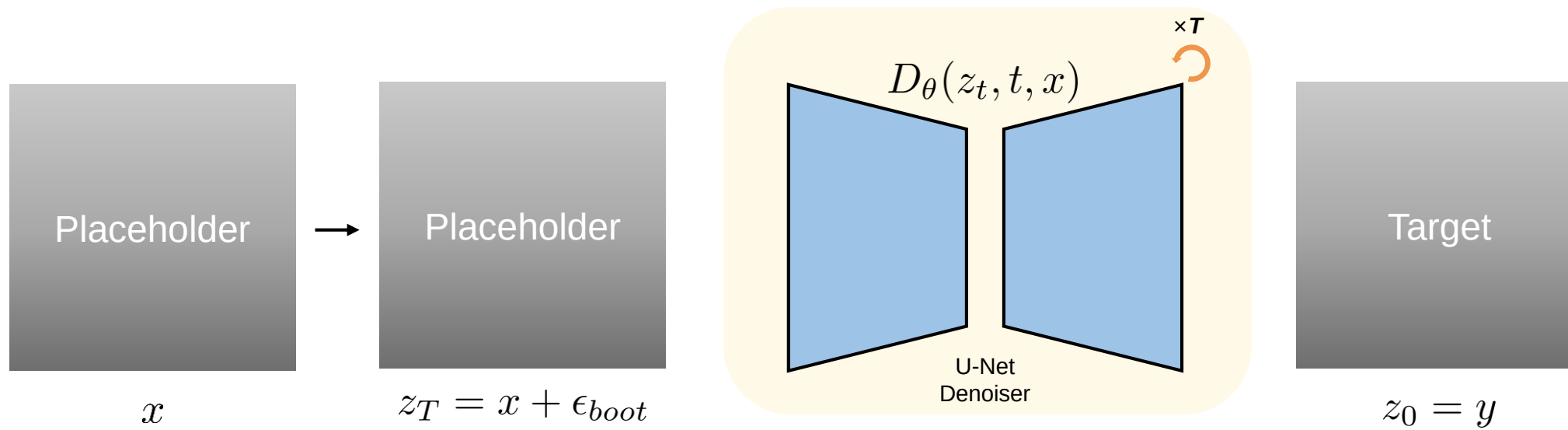
Place-  
holder

Place-  
holder

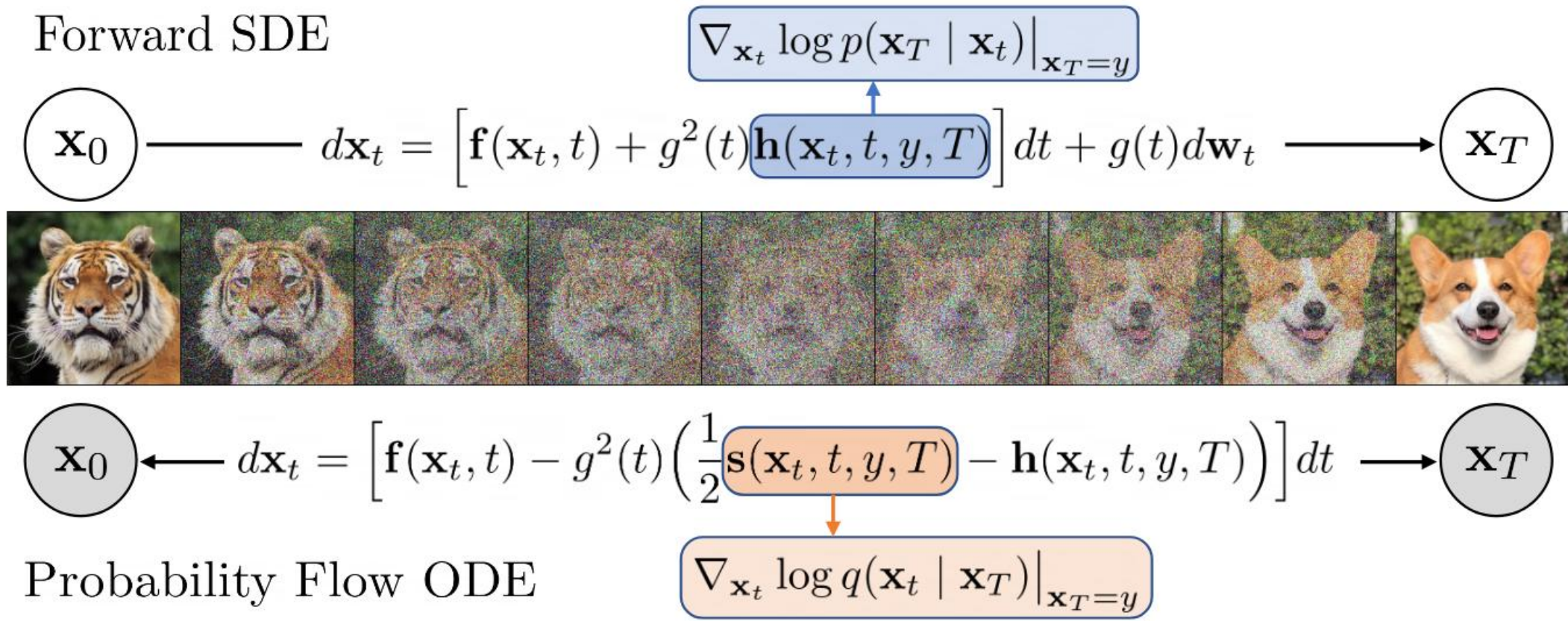
Place-  
holder











**Figure 1: A schematic for Denoising Diffusion Bridge Models.** DDBM uses a diffusion process guided by a drift adjustment (in blue) towards an endpoint  $\mathbf{x}_T = y$ . They learn to reverse such a bridge process by matching the denoising bridge score (in orange), which allows one to reverse from  $\mathbf{x}_T$  to  $\mathbf{x}_0$  for any  $\mathbf{x}_T = y \sim q_{\text{data}}(y)$ . The forward SDE process shown on the top is unidirectional while the probability flow ODE shown at the bottom is deterministic and bidirectional. White nodes are stochastic while grey nodes are deterministic.